

$$\dot{Q} = \dot{m}c_p\Delta T + mc_p\Delta T$$

$$A = A_0 e^{-\lambda t}$$

$$\dot{Q} = \dot{m}\Delta h$$

$$P = P_0 e^{t/\tau}$$

$$P = P_0 10^{SUR(t)}$$

$$\dot{W}_{bomba} = \dot{m}v\Delta P = \dot{m}\Delta P/\rho$$

$$CR_1(1 - K_{ef_1}) = CR_2(1 - K_{ef_2})$$

$$\dot{m} = \rho A \vec{v} = \rho \dot{v}$$

$$1/M = CR_1/CR_X$$

$$N = S/(1 - K_{ef})$$

$$F = PA \quad A = \pi r^2$$

$$\rho = \frac{K_f - K_i}{K_f K_i}$$

$$1/M = 1 - K_{ef}$$

$$\dot{v} = k\sqrt{\Delta P}$$

$$\rho = (K_{ef} - 1)/K_{ef}$$

$$C_{f,B} = \frac{\Delta \rho}{\alpha_B} + C_{i,B}$$

$$\dot{Q} \propto \dot{m}_{Circ Nat}^3$$

$$K_{ef} = 1/(1 - \rho)$$

$$\Delta T \propto \dot{m}_{Circ Nat}^2$$

$$SUR = 26,06/\tau \quad (\tau \text{ em s})$$

$$\text{Bernoulli: } z + \frac{\vec{v}^2}{2g} + \frac{p}{\rho g} + H = \text{cte}$$

$$T_R = \Phi \Sigma$$

$$\Sigma = \sigma N$$

$$\dot{Q} = UA\Delta T$$

$$\tau = \frac{\bar{\beta}_{ef} - \rho}{\rho \lambda_{ef} + \dot{\rho}}$$

$$\rho = \frac{\ell^*}{\tau} + \frac{\bar{\beta}_{ef}}{1 + \tau \lambda_{ef}}$$

$$K = ^\circ\text{C} + 273,15$$

$$\ell^* = 10^{-4} \text{ s}$$

$$\alpha_x = \frac{\Delta \rho}{\Delta X}$$

$$P = VI \quad P_{aparente} = \sqrt{3}VI$$

$$\lambda_{ef} = 0,1 \text{ s}^{-1} \quad (\text{para } \rho \text{ pequena e positiva})$$

$$P_{ativa} = \sqrt{3}VI \cos \varphi = \sqrt{3}VI f_p$$

$$\text{Valor Diferencial Reatividade} = k \frac{\varphi_{ponta}^2}{\varphi_{médio}^2}$$

$$P_{reativa} = \sqrt{3}VI \sin \varphi$$

$$\text{Eficiência} = \frac{\text{Trabalho Líquido Saída}}{\text{Energia Entrada}}$$

$$X_L = 2\pi fL \quad X_C = \frac{1}{2\pi fC}$$

$$v_i = k(T_M - T_{Ref})$$

$$v_i \uparrow \quad T_i \uparrow$$

$$H = h_a + h_r + J_a + J_r + \frac{(v_r^2 - v_a^2)}{2g}$$

$$NPSH_d = H_b - h_a - h_v - J_a$$

$$P_{\text{Pico}} = \rho \cdot C_{\text{Som}} \cdot \Delta \vec{v}$$

$$\dot{W} = \frac{1000.Q.H}{75.\eta} \text{ (CV)}$$

$$P_{\text{máx}} = P_{\text{Estática}} + P_{\text{Pico}}$$

$$Acumulação = \frac{P_{\text{máx}} - \textit{Setpoint}}{\textit{Setpoint}}$$

$$\text{Diferencial Alívio (blowdown)} = \frac{\textit{Setpoint} - \textit{Reseat}}{\textit{Setpoint}}$$

Leis bomba centrífuga fluxo radial:

$$\frac{\dot{V}}{\dot{V}_1} = \frac{n}{n_1} \qquad \frac{H}{H_1} = \left(\frac{n}{n_1}\right)^2 \qquad \frac{\dot{W}}{\dot{W}_1} = \left(\frac{n}{n_1}\right)^3 \qquad I = k\dot{V}^3$$

$$\frac{\dot{V}}{\dot{V}_1} = \frac{D}{D_1} \qquad \frac{H}{H_1} = \left(\frac{D}{D_1}\right)^2 \qquad \frac{\dot{W}}{\dot{W}_1} = \left(\frac{D}{D_1}\right)^3$$